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Title: Self-Orienting Stabilization of a High-Altitude Balloon Payload using IMU Feedback

Abstract:

This project develops a stabilization system for a high-altitude balloon payload that maintains a desired orientation relative to Earth. An inertial measurement unit (IMU) measures the payload's yaw angle, and the system uses this feedback to command a servo motor that rotates the payload to counteract disturbances. This experiment demonstrates how real-time sensor feedback can be used to actively correct payload orientation and improve stability during flight.